

Dataset Description

1. Introduction

The dataset used in this project consists of chest X-ray images collected for automated detection and localization of thoracic abnormalities. Medical imaging datasets are widely used in artificial intelligence research to assist in disease diagnosis and improve clinical decision-making. The dataset contains radiographic images along with annotation data provided by expert radiologists, making it suitable for computer vision tasks such as object detection and classification.

2. Dataset Preparation for Milestone-1

Due to the large size of the original dataset, a smaller subset of images was selected for preprocessing and experimentation. A total of 15 DICOM images were used locally for testing the preprocessing pipeline. The annotation file originally contained over 67,000 rows, representing multiple bounding box annotations per image. The dataset was filtered so that only rows corresponding to the selected images were retained, resulting in a smaller annotation file with relevant bounding boxes.

3. Data Format and Components

3.1 Image Data

- Image Type: Chest X-ray radiographs
- Format: DICOM (.dicom)
- Color Mode: Grayscale
- Resolution: High-resolution medical images

DICOM files contain both pixel data and metadata related to imaging parameters. These images must be converted into standard image formats such as PNG before applying computer vision techniques.

3.2 Annotation Data

The dataset includes a metadata file containing bounding box annotations. Each row corresponds to a single object instance within an image. Important fields include:

- `image_id` — Unique identifier for each image
- `class_name` — Name of the abnormality
- `class_id` — Numeric class label
- `x_min`, `y_min`, `x_max`, `y_max` — Bounding box coordinates

Images labeled as “No Finding” do not contain bounding boxes and represent healthy cases.

3.3 Data Characteristics

Key characteristics observed during dataset exploration include:

- Multiple annotations may exist for a single image.
- Bounding box coordinates vary widely due to differences in image resolution.
- Some images contain no abnormalities, resulting in empty bounding box lists.
- Pixel intensity values vary significantly because of medical imaging conditions.

4. Dataset Challenges

Medical imaging datasets introduce several technical challenges:

- Large image dimensions requiring resizing.
- High dynamic range pixel intensities needing normalization.
- Variability among radiologist annotations.
- Class imbalance between healthy and abnormal cases.

These challenges require careful preprocessing before training machine learning models.

5. Dataset Exploration

Initial exploration of the dataset involved:

- Inspecting annotation distributions using the CSV file.
- Loading sample DICOM images to understand pixel structure.
- Converting selected images into PNG format for visualization.

This exploratory process helped determine the preprocessing steps required to standardize the data.

6. Conclusion

The dataset provides a rich collection of annotated chest X-ray images suitable for building AI-based detection systems. By filtering the annotation file and selecting a subset of images, a manageable dataset was created for preprocessing and experimentation. This prepared dataset forms the foundation for further stages of the project, including model training and evaluation.